Chapter 3

Simple and Decimal Fractions

A digit which can be represented in p/q form, where $q \neq 0$ is called a fraction. Here, p is called the numerator and q is called the denominator.
For example 3/5 is a fraction, where 3 is called numerator and 5 is called denominator.

OR

When a unit is divided into any number of equal parts, then these parts are termed as a fraction of the unit.
For example: If 1 is to be divided into two equal parts, then 1 is divided by 2, and is represented as $\frac{1}{2}$

3.1 Simple Fraction

The fraction which has denominator other than power of 10 is called simple fraction. *For example:* $\frac{3}{7}, \frac{5}{11}, \frac{7}{9}$ etc.

Note: Simple fraction is also known as **vulgar fraction.**

Types of Simple Fractions

There are following types of fractions.

1. **Proper Fraction:** When the numerator of a fraction is less than its denominator, then fraction is called proper fraction.
   For example: $\frac{1}{2}, \frac{15}{17}, \frac{21}{43}, \ldots$
2. **Improper Fraction:** When the numerator of a fraction is greater than its denominator, then fraction is called improper fraction.
   For example: $\frac{17}{13}, \frac{18}{14}, \frac{45}{19}, \ldots$
3. **Compound Fraction:** A fraction, in which numerator or denominator or both are in fraction, then it is called compound fraction.
   For example: $\frac{11}{7/9}, \frac{11/9}{13}, \frac{1/4}{7/14}, \ldots$
4. **Inverse Fraction:** If we inverse the numerator and the denominator of a fraction, then the resultant fraction will be the inverse fraction of the original fraction.
   For example: Given fraction $= \frac{3}{8}$; Inverse fraction $= \frac{8}{3}$
5. **Mixed Fraction:** The fraction, which is the combination of integer and fraction, is called mixed fraction.
   For example: $3\frac{2}{5}, 7\frac{1}{9}, \ldots$
6. **Continuous Fraction:** It has no certain definition but only say that a fraction contains additional fractions in its denominators, is called continuous fraction.
   For example:

$$(i) \quad 2 + \frac{1}{2 + \frac{2}{5+\frac{2}{3}}} \qquad (ii) \quad 6 + \frac{1}{1 + \frac{1}{\frac{1}{2}+\frac{1}{2}}}$$

Note: To simplify a continuous fraction, start from bottom and work upwards

3.2 Decimal Fraction

If the fraction has denominator in the powers of 10, then fraction is called decimal fraction.
For example: 10th part of unit $= \frac{1}{10} = 0.1$; 10th part of $6 = \frac{6}{10} = 0.6$

To convert a decimal fraction into a vulgar fraction, place 1 in the denominator under the decimal point. Then, after removing the decimal point, place as many zeroes after it as the number of digits after the decimal point. Finally, reduce the fraction to its lowest terms.

For example: $0.23 = \frac{0.23}{1} = \frac{23}{100}$; $0.0035 = \frac{0.0035}{1} = \frac{35}{10000} = \frac{7}{2000}$

Note:

- ✦ Placing zeroes to the right of a decimal fraction, it does not make any change in value. Hence, 0.5, 0.50, 0.500 and 0.5000 are equal.
- ✦ If numerator and denominator of a fraction have same number of decimal places, then each of the decimal points be removed.

  For example: $\dfrac{0.0005}{0.0017} = \dfrac{5}{17}$

Types of Decimal Fractions

1. **Recurring Decimal Fraction:** The decimal fraction, in which one or more decimal digits are repeated again and again, is called recurring decimal fraction. To represent these fractions, a line is drawn on the digits which are repeated.
   For example: $\dfrac{2}{3} = 0.666\cdots = 0.\overline{6}$; $\quad \dfrac{22}{7} = 3.142857142857\cdots = 3.\overline{142857}$
2. **Pure Recurring Decimal Fraction:** When all the digits in decimal fraction are repeated after the decimal point, then the decimal fraction is called as pure recurring decimal fraction.
   For example: $0.\overline{5}, 0.\overline{489}, \ldots$
   To convert pure recurring decimal fractions into simple fractions (vulgar form), write down the repeated digits only once in numerator and place as many nines in the denominator as the number of digits repeated.
   For example:

   (a) $0.\overline{3} = \dfrac{3}{9} = \dfrac{1}{3}$
   Since, there is only 1 repeated digit. Therefore, only single 9 is placed in denominator.

   (b) $0.\overline{57} = \dfrac{57}{99} = \dfrac{19}{33}$
   Since, there are only 2 repeated digits. Therefore, two 9's are placed in denominator.
3. **Mixed Recurring Decimal Fraction:** A decimal fraction in which some digits are repeated and some are not repeated after decimal is called as mixed recurring decimal fraction. *For example:* $3.2\overline{23}, 0.1\overline{9}$
   To convert mixed recurring decimal fractions into simple fractions, in the

numerator, take thedifference between the number formed by all the digits after decimal point (repeated digits will be taken only once) and the number formed by non-repeating digits. In the denominator, place as many nines as there are repeating digits and after nine put as many zeroes as the number of non-repeating digits.
For example:

(a) $0.3\overline{6} = \dfrac{36-3}{9} = \dfrac{33}{99} = \dfrac{11}{9}$

(b) $0.4\overline{267} = \dfrac{4267-4}{9990} = \dfrac{4263}{9990} = \dfrac{1421}{3330}$

3.3 Important Facts Related to Simple and Decimal Fractions

- If in a fraction, numerator is equal to denominator, then the value of fraction is equal to 1.*i.e,* $\dfrac{3}{3} = \dfrac{11}{11} = 1$
- If the numerator of a fraction is always non-zero and denominator is zero, then the value of fraction is infinity. *i.e,* $\dfrac{5}{0} = \dfrac{3}{0} = \infty$.
- If the numerator of a fraction is zero and denominator is not equal to zero, then the value of fraction is zero.*i.e,* $\dfrac{0}{11} = \dfrac{0}{99} = 0$
- If the numerator or denominator of any fraction is either multiplied or divided by same number, then the value of fraction remains unchanged.
  i.e, $\dfrac{3}{5} = \dfrac{6}{10} = \dfrac{9}{15}, \quad \dfrac{1}{2} = \dfrac{2}{4} = \dfrac{5}{10}$
- If the numerator and denominator have no common factor other than 1, then the fraction is said to be in its lowest form.*i.e,* $\dfrac{1}{2}, \dfrac{3}{4}$ are in lowest form but $\dfrac{2}{4}, \dfrac{6}{10}$ are not.
- A fraction is a rational number as it can be expressed in the form of p/q and $q \neq 0$

3.4 Operations on Simple Fractions

Addition of Simple Fractions

1. *When Denominators are Same:* If denominators of fractions are same, then numerators of fractions are added and their addition is divided by denominator.*For example:*

$$\frac{1}{4} + \frac{2}{4} = \frac{1+2}{4} = \frac{3}{4}$$

2. *When Denominators are Different:* If denominators of fractions are not same, then make their denominators equal (by taking their LCM) and then add their numerators.*For example:*

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{6}{12} + \frac{4}{12} + \frac{3}{12} = \frac{6+4+3}{12} = \frac{13}{12}$$

Subtraction of Simple Fractions

1. *When Denominators are Same:* If denominators of fractions are same, then numerators of fractions are subtracted and their subtraction is divided by the denominator. *For example:*

$$\frac{3}{4} - \frac{1}{4} = \frac{3-1}{4} = \frac{2}{4} = \frac{1}{2}$$

2. *When Denominators are Different:* If denominators of fractions are not same, then make their denominators equal and then subtract their numerators.*For example:*

$$\frac{2}{3} - \frac{1}{2} = \frac{(2\times 2) - (3\times 1)}{2\times 3} = \frac{4-3}{6} = \frac{1}{6}$$

Multiplication of Simple Fractions

To multiply two or more simple fractions, multiply their numerators and denominators. *For example:*

$$\frac{1}{2} \times \frac{3}{4} = \frac{1\times 3}{2\times 4} = \frac{3}{8}$$

If fractions are given in mixed form, first convert them into improper fraction and then multiply. *For example:*

$$2\frac{4}{5} \times 1\frac{8}{3} = \frac{14}{5} \times \frac{11}{3} = \frac{14\times 11}{5\times 3} = \frac{154}{15}$$

Division of Simple Fractions

To divide two fractions, first fraction is multiplied by the inverse of second fraction.
For example:

$$\frac{2}{3} \div \frac{3}{5} = \frac{2}{3} \times \frac{5}{3} = \frac{2 \times 5}{3 \times 3} = \frac{10}{9}$$

3.5 Operations on Decimal Fractions

Addition and Subtraction of Decimal Fractions

To add or subtract decimal fractions, the given numbers are written under each other such that the decimal points lie in one column and the numbers so arranged can now be added or subtracted as per the conventional method of addition and subtraction.
Ex. 1 (i) 353.5 + 2.32 + 43.23 =? (ii) 1000 − 132.23 =?
Sol.

$$(i) \quad \begin{array}{r} 353.50 \\ + \quad 2.32 \\ 43.23 \\ \hline 399.05 \end{array} \qquad (ii) \quad \begin{array}{r} 1000.00 \\ - \quad 132.23 \\ \hline 867.77 \end{array}$$

Multiplication of Two or More Decimal Fractions

Given fractions are multiplied without considering the decimal points and then in the product, decimal point is marked from the right hand side to as many places of decimal as the sum of the numbers of decimal places in the multiplier and the multiplicand together.
Ex. 2 (i)4.3×0.13 =? (ii) $1.12 \times 2.3 \times 4.325$ =?
Sol. (i)

$$43 \times 13 = 559$$

Sum of the decimal places $= (1 + 2) = 3$

$\therefore$ Required product $= 0.559$

(ii)

$$112 \times 23 \times 4325 = 11141200$$

Sum of the decimal places $= (2 + 1 + 3) = 6$

$\therefore$ Required product $= 11.141200$

Multiplication of Decimal Fraction by an Integer

Given integer is multiplied by the fraction without considering the decimal point and then in the product, decimal is marked as many places before as that in the given decimal fraction.

Ex. 3 Find the value of the following.

(i) 19.72×4 (ii) 0.0745×10 (iii) 3.52×14

Sol. (i) 19.72×4

Multiplying without taking decimal point into consideration

$$1972 \times 4 = 7888$$

So, $19.72 \times 4 = 78.88$

Since, in the given decimal fraction, decimal point is two places before. So, in the product, decimal point will also be put two places before. Similarly,

(ii) $0.0745 \times 10 = 0.745$

(iii) $3.52 \times 14 = 49.28$

Dividing a Decimal Fraction by an Integer

Do simple division i.e., divide the given decimal number without considering the decimal point and place the decimal point as many places of decimal as in the dividend.

Ex. 4 Divide the following.

(i)$0.81 \div 9$ (ii)$1.2875 \div 25$ (iii) $0.00049 \div 7$

Sol.

(i) $\frac{81}{9} = 9 \implies \frac{0.81}{9} = 0.09$ [Two places of decimal]

(ii) $\frac{12875}{25} = 515 \implies \frac{1.2875}{25} = 0.0515$ [Four places of decimal]

(iii) $\frac{49}{7} = 7 \implies \frac{0.00049}{7} = 0.00007$ [Five places of decimal]

Division of Decimal Fractions

In such divisions, dividend and divisor both are multiplied first by a suitable multiple of 10 to convert divisor into a whole number and then above mentioned rule of division is followed.

Ex. 5 Divide the following.

(i) $42 \div 0.007$ (ii) $0.00048 \div 0.8$

Sol.

(i) $$\frac{42}{0.007} = \frac{42}{0.007} \times \frac{1000}{1000} = \frac{42000}{7} = 6000$$

(ii) $$\frac{0.00048}{0.8} = \frac{0.00048}{0.8} \times \frac{10}{10} = \frac{0.0048}{8} = 0.0006$$

3.6 Comparison of Simple Fractions

Following are some techniques to compare fractions.

Cross Multiplication Method

If $\frac{a}{b}$ and $\frac{c}{d}$ are two fractions. Then

1. If $ad > bc$, then $\frac{a}{b} > \frac{c}{d}$
2. If $ad < bc$, then $\frac{a}{b} < \frac{c}{d}$
3. If $ad = bc$, then $\frac{a}{b} = \frac{c}{d}$

Ex. 6 (i)Between $\frac{4}{7}, \frac{3}{8}$, which fraction is bigger?

(ii) Which one of the fractions is the largest among $\frac{2}{3}, \frac{3}{4}, \frac{4}{3}, \frac{5}{4}$

Sol. (i) Given $\frac{4}{7}$ and $\frac{3}{8}$

$\because \quad 4 \times 8 > 3 \times 7$

$\therefore \quad \frac{4}{7} > \frac{3}{8}$

(ii) First find the largest among the two fractions

$$\frac{2}{3}, \frac{3}{4} \quad ; \quad \frac{4}{3}, \frac{5}{4}$$

$\because \quad 3 \times 3 > 3 \times 4 \quad ; \quad 4 \times 4 > 5 \times 3$

$\therefore \quad \frac{3}{4} > \frac{2}{3} \quad ; \quad \frac{4}{3} > \frac{5}{4}$

Now taking the largest two fractions $\frac{3}{4}$ and $\frac{4}{3}$

$\because \quad 4 \times 4 > 3 \times 3$

$\therefore \quad \frac{4}{3} > \frac{3}{4}$

$\therefore \quad \frac{4}{3}$ is the largest fraction.

By Changing Fractions in Decimal Form

To compare two or more fractions, first convert fractions into decimal form and then compare.

Ex. 7 Between $\frac{1}{7}$ and $\frac{2}{9}$, which fraction is bigger?

Sol. $\frac{1}{7} = 0.14$ and $\frac{2}{9} = 0.22$

It is clear that $0.22 > 0.14$

$\therefore \quad \frac{2}{9} > \frac{1}{7}$

Hence, the bigger fraction is $\frac{2}{9}$

By Equating Denominators of Given Fractions

For comparison of fractions, take LCM of the denominators of all fractions, so that the denominators of all fractions are same. Now, the fraction having largest numerator is the largest fraction.

Ex.8 Arrange the following fractions in decreasing order $\frac{3}{5}, \frac{7}{9}, \frac{11}{13}$.

Sol. LCM of 5,9,13 = $5 \times 9 \times 13$

$\because$ $\frac{3}{5} = \frac{3 \times 9 \times 13}{5 \times 9 \times 13} = \frac{351}{585}$,

$\frac{7}{9} = \frac{7 \times 5 \times 13}{9 \times 5 \times 13} = \frac{455}{585}$

and $\frac{11}{13} = \frac{11 \times 5 \times 9}{11 \times 5 \times 9} = \frac{495}{585}$

Now the fraction with largest numerator will be the largest.

$\therefore$ Decreasing order will be $\frac{495}{585}, \frac{455}{585}, \frac{351}{585}$

Hence, the order is $\frac{11}{13}, \frac{7}{9}, \frac{3}{5}$

By Equating Numerators of Given Fractions

For comparison of fractions, take LCM of the numerator of all fractions, so that numerators of all the fractions are same. Now, the fraction having smallest numerator will be largest.

Ex. 9 Which fraction is largest among $\frac{3}{13}, \frac{2}{15}, \frac{4}{17}$?

Sol. LCM of 2,3,4 = $2 \times 2 \times 3 = 12$

$\because$ $\frac{3}{13} = \frac{3 \times 4}{13 \times 4} = \frac{12}{52}, \frac{2 \times 6}{15 \times 6} = \frac{12}{90}$ and $\frac{4}{17} = \frac{4 \times 3}{17 \times 3} = \frac{12}{51}$

Now, fraction having the smallest denominator will be largest.

Hence $\frac{4}{17}$ is the largest.

3.7 Fast Track Formulae to solve the QUESTIONS

Formula 1

To represent any fraction in simplified form, divide its numerator and denominator by their HCF.

Ex. 10 Write $\frac{27}{81}$ in simplified form.

Sol. $\because$ $27 = 3 \times 3$ and $81 = 3 \times 3 \times 3 \times 3$

HCF of 27 and 81 $= 3 \times 3 \times 3$

$\therefore$ $\frac{27}{81} = \frac{1 \times 27}{3 \times 27} = \frac{1}{3}$

Formula 2

If in the given fractions, the difference between numerator and denominator are same, then fraction having larger numerator is the largest and fraction having smaller numerator is the smallest.

Ex. 11 Arrange the given fractions in increasing order, $\frac{4}{5}, \frac{5}{6}, \frac{6}{7}$

Sol. Since all the fractions have difference in numerator and denominator the same.

$\therefore$ Increasing order is $\frac{4}{5}, \frac{5}{6}, \frac{6}{7}$

Formula 3

If in the given fractions, the numerators are increasing by a definite value and the denominator is also increasing by a definite value but the value of denominator is greater than numerator, then the fraction having smaller numerator will be the smallest fraction and the fraction having larger numerator will be the largest fraction.

Ex. 12 Which of the fractions is largest among $\frac{2}{5}, \frac{5}{11}, \frac{8}{17}, \frac{11}{23}$?

Sol. Since, in the given fractions, numerator value is increasing by 3 and denominator value is increasing by 6 and $6 > 3$. Then, the fraction having larger numerator will be the larger fraction.

$\therefore$ $\frac{11}{23}$ is the largest fraction.

Formula 4

If any number is divided by a/b instead of multiplying by a/b, then the obtained value will be x greater than original value then,

$$\text{The given number will be } \frac{abx}{b^2 - a^2}$$

Ex. 13 Arun was to find 6/7 of a fraction. Instead of multiplying, he divided the fraction by 6/7 and the result obtained was 13/70 greater than original value. Find the fraction given to Arun?

Sol. Given, $a = 6$, $b = 7$, and $x = \dfrac{13}{70}$

$$\therefore \quad \text{Required number } = \frac{abx}{b^2 - a^2} = \frac{6 \times 7 \times \frac{13}{70}}{7^2 - 6^2} = \frac{6 \times 13}{10 \times 13} = \frac{3}{5}$$

$$\therefore \quad \text{Fraction given to Arun is } \frac{3}{5}$$